S1: Introduction to Social Media and Text Analytics

Agenda

- Why learn social media and text analytics?
- What we will learn?
- Course format
- Expectation

Why I need to learn another data analytics course?

- What kinds of data you analyzed in previous analytics/math/stats courses?

Sales Manager [Sales-Dive A]					
Sales Manager	**Count**	**Bottles**	**Cases**	**Cost**	**Revenue**
Totals	2,686,941	4,694,652	4,506,420	430,348,302.37	498,932,959.70
Benett, Keith [205]	224842	408151	346983	36794340.90	42242652.28
Bowen, Patricia [211]	344027	580202	595974	[illegible]	[illegible]
Clarkson, Christopher [215]	349526	576724	621715	[illegible]	[illegible]
Donnelly, Jeanne [201]	283983	491941	471661	[illegible]	[illegible]
Fermier, Raymond [208]	321837	571152	537117	[illegible]	[illegible]
Jackson, Lisa [217]	332608	604662	504834	[illegible]	[illegible]
Lewis, Bernard [204]	229641	409982	393453	[illegible]	[illegible]
Norton, Allan [202]	330917	581310	575801	56052791.16	64953334.89
Peterson, David [210]	269560	470528	458882	48805365.22	56463822.10

Product master data

Product ID	Product name	Product description	Price
1982SP	Lamp	Metal floor-standing lamp	56.95
454YH	Chair	Oak dining chair	70.99

Customer master data

Customer ID	Customer name	Customer address
67	J. Smith	56 Quy Road, Chester, UK
68	R. Hurst	14 Back Lane, Norwich, UK

Transactions

Sale ID	Product ID	Customer ID	Actual Sale price	Sale time
002	1982SP	67	56.95	1/3/13 15.34.12
003	454YH	67	65.99	1/3/13 15.34.25
004	454YH	68	70.99	4/3/13 12.05.43

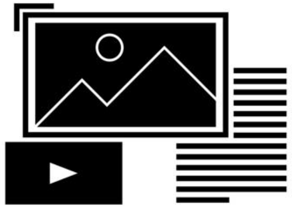
What kinds of data you will face in business?

Modern use of data

- Monitor sentiment
- Identify trend
- Identify influencers
- Segment customer
- …..

Their data source and format:

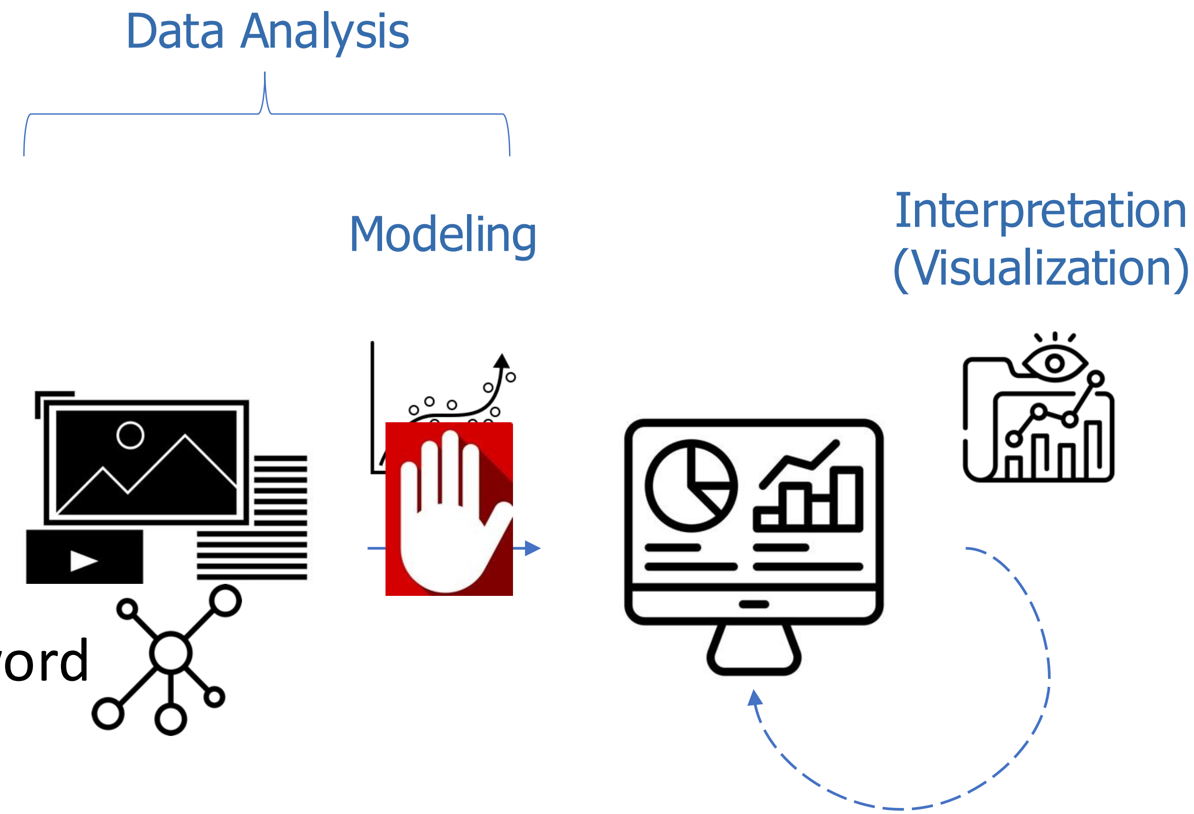
- Social media data
- Text
- Image
- Networks



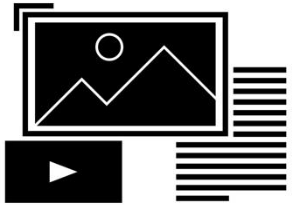
Can you analyze these data just like the tabular data?

\$1 + \$-0.5

JZ is awesome, a nice teacher +
DJZ is aweful, I cannot understand a single word



Structured vs. unstructured data

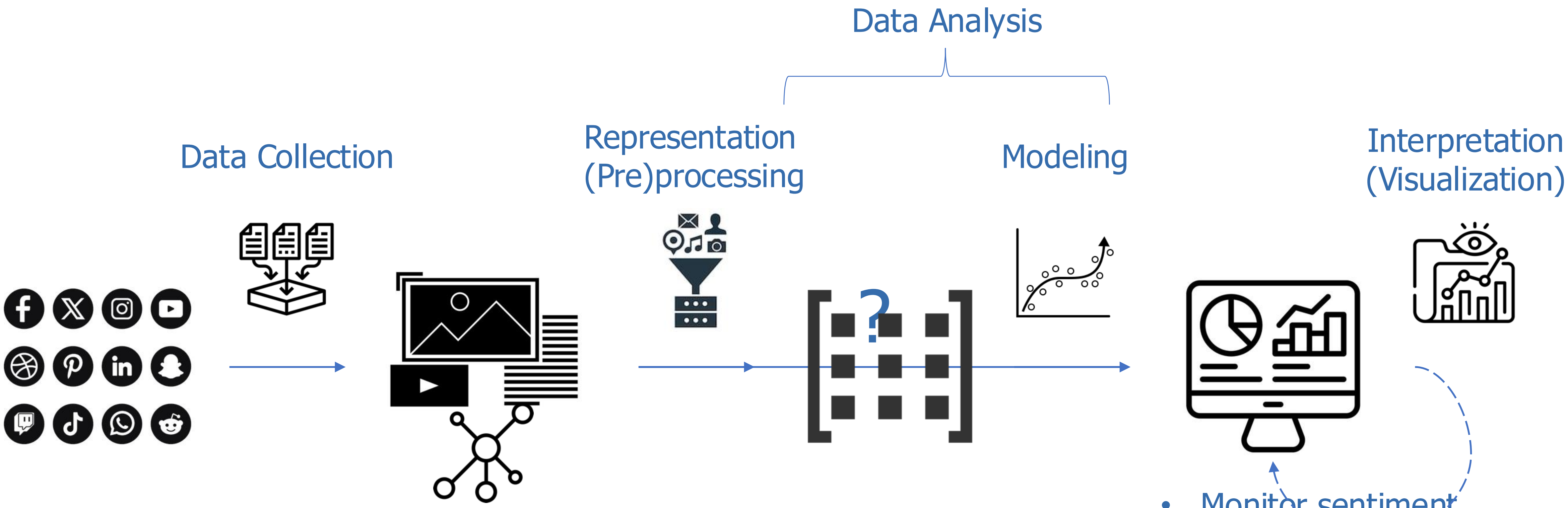


- Unstructured data
  - No pre-defined data model

  - Ambiguous relationships and elements
  - Example: email, video, and ?
  - Pro: low cost to generate and curate
  - Con: high cost to extract information, difficult to analyze
  - More than 80%
  - Soon 99%

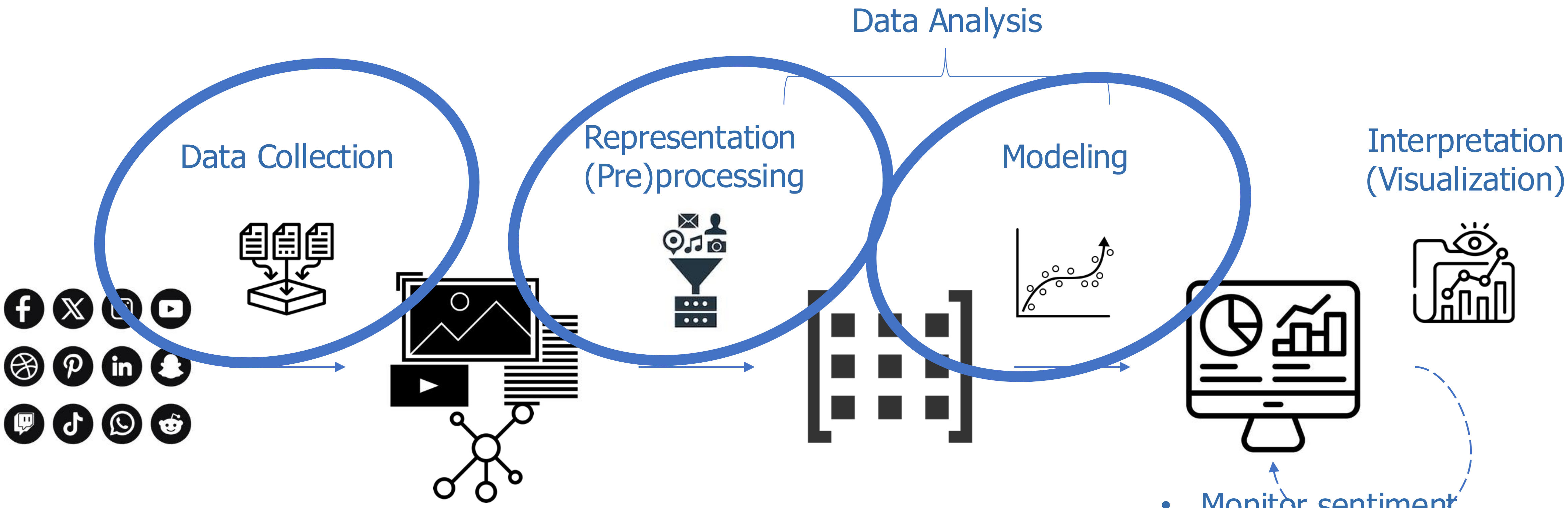
- Structured data
  - Pre-defined, (ordered, organized, structured) data model
  - Clear relationships between elements
  - Example: spread sheet, relational database, and ?
  - Pro: low cost to extract information, ready to analyze
  - Con: high cost to generate, collect/curate
  - Less than 20%

How to analyze social media (unstructured) data?



- Monitor sentiment
- Monitor topic
- Identify trend
- Identify influencers
- Segment customer

What will we learn?



- Social networks
- Text data

- Monitor sentiment
- Monitor topic
- Identify trend
- Identify influencers
- Segment customer

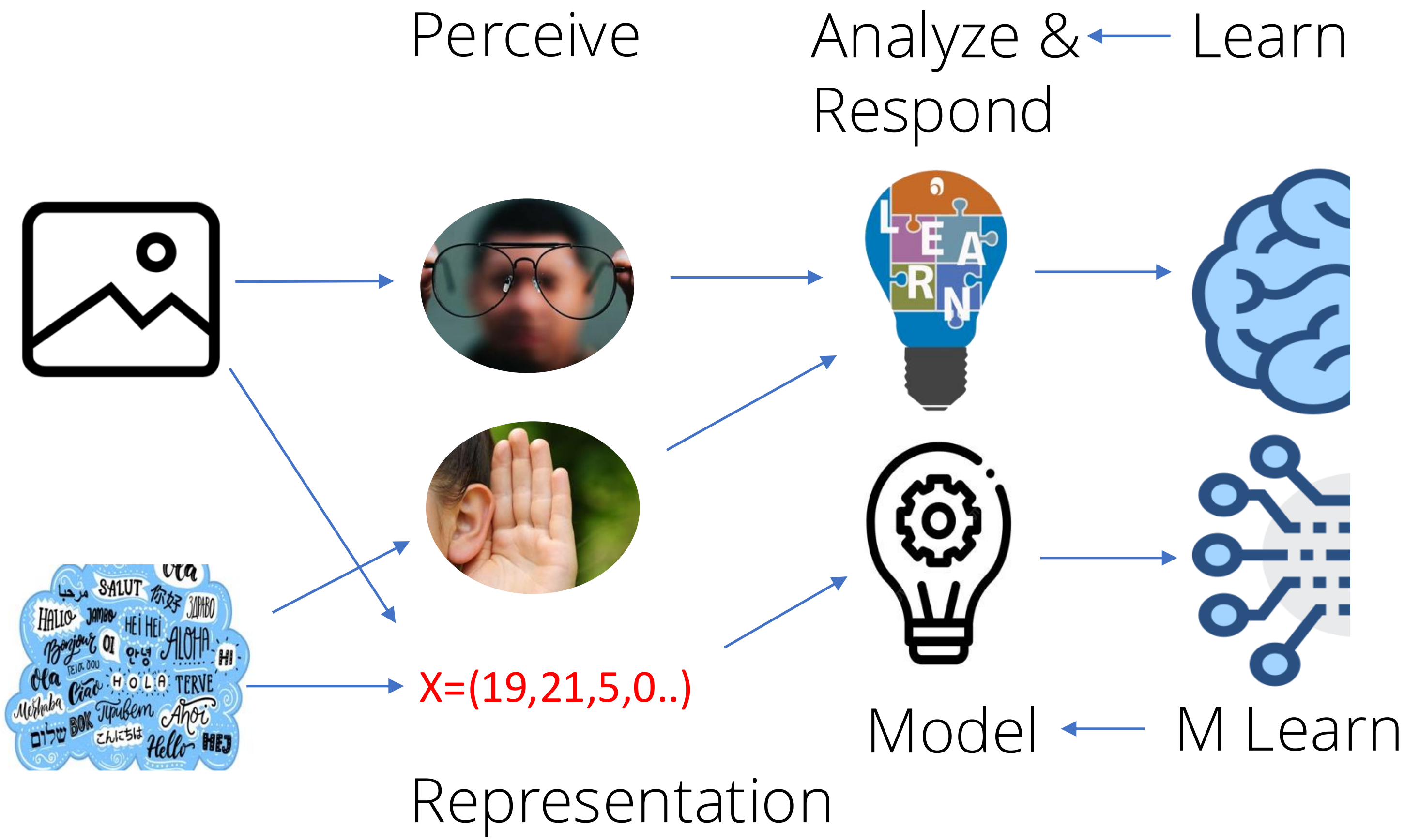
Syllabus

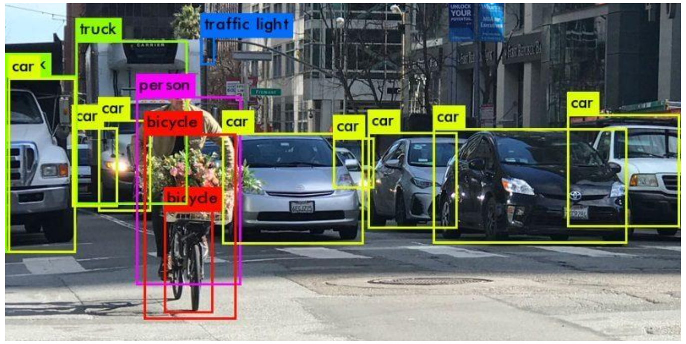
Session		Topic
1	17/18 Mar	Course Introduction
Module 1: Web/Text Data Collection		
2	Asyn 1	Scrape One Webpage
3	Asyn 2	Specifying XPath
4	Asyn 3	Scrape Hierarchical Features
5	Asyn 4	Scrape with Pagination
6	Asyn 5	Scrape Hierarchical Pages I
7	Asyn 6	Scrape Hierarchical Pages II
Module 2: Text Analysis		
8	19/20 Mar	Text Representation I: General
9	24/25 Mar	Text Representation II: Document/Sentence Embedding
10	26/27 Mar	Unsupervised Learning: Text Similarity & Clustering
11	31 Mar/1 Apr	Unsupervised Learning: LDA Topic Model & Sentiment Analysis
12	2/3 Apr	Sentiment Analysis for Text
13	7/8 Apr	Supervised Learning for Text
14	9/10 Apr	Text Representation III: Word Embedding
15	14/15 Apr	Deep Learning for Text (RNN & LSTM)
16	16/17 Apr	Sequence to Sequence Model
Module 3: Social Network Analysis		
17	21/22 Apr	Network Basic and Representation
18	23/24 Apr	Network Diagnosis
19	28/29 Apr	Identify Influencers
20	Asyn 7	Detect Communities
21	30 Apr/1 May	Roadshow day: Group Project Presentations

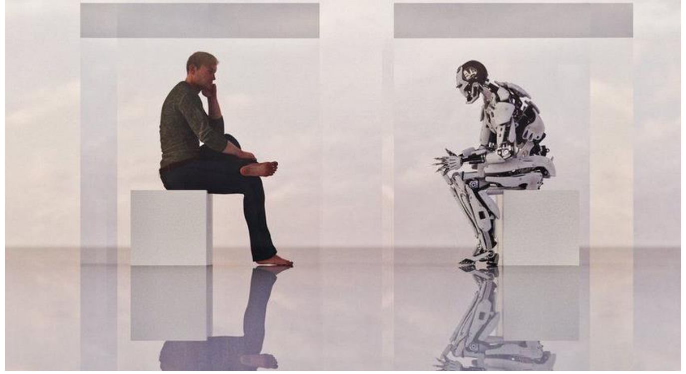
Beyond social media context

- M1: Scrape any web data
  - Structured data
- M2: Analyze any text and unstructured data
  - Natural language processing (NLP)
  - LLMs, e.g., chatbot
  - Machine learning & AI
- M3: Analyze other networks
  - Transportation networks
  - Public health, e.g., disease transmission
  - Disaster response, e.g., resource distribution
  - Crime networks
  - Policy communication

Overlap with advanced machine learning and AI







Class format

- Mechanical works
- **Identify (vs. solve) a question**
- **Systematic planning**
- Focus: conceptual
- Coding: GenAI allowed

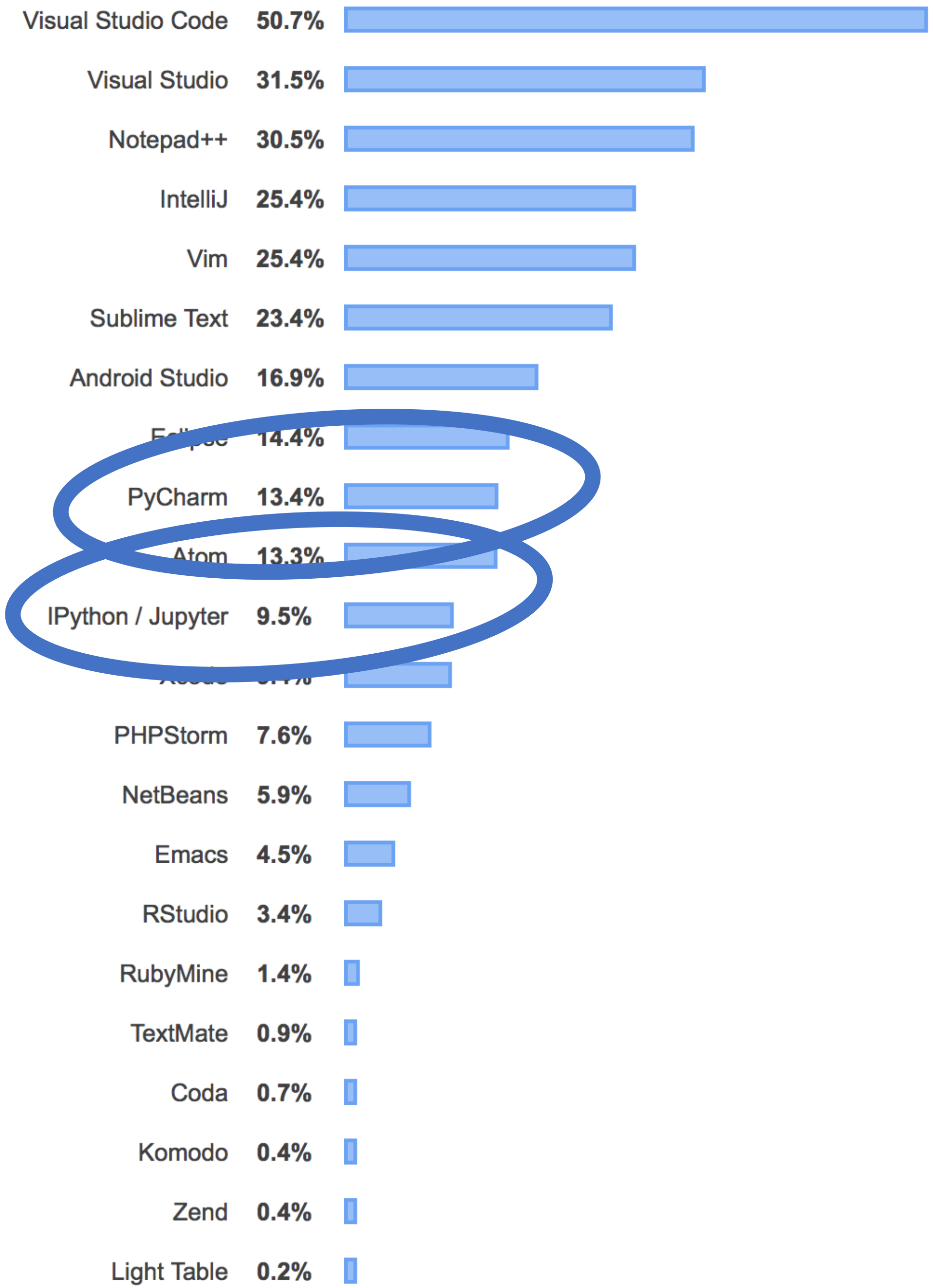
Experiential learning: Interactive, conversational, fun, real-life BA

- Wa-rm(-ke)-up quiz
- Seminar 1/3
- Implementation 1/3
- Mini project 1/3, peer-review to debug, AI assistance, attention
- No individual assignment

Environment

- BYOD!!
- Core: Python 3
- IDE: PyCharm Community Version (free)

Eclipse PyDev Sublime Text Atom Emacs Vim



Quizzes (35pt)

1 cumulative quiz

- 35pt
- In-class, closed book
- 60 mins to 90 mins
- No AI
- Industrial technical interview questions

Warm-up quizzes

- Every session
- Usually finish in classes; Otherwise, due at the end (23:59) of the next day to the class
- Not to grade
- Count toward participation. You get points by doing, even wrongly. Incorrect solutions will be appreciated more!!
- No AI

Group projects (55pt)

Course Project

- Industry-level, focusing on M3
- 25pt (milestone 1 + code + report + lightning talk)
- Class size/15 members (form at the 2nd week, FCFS), self-enrolled
- Project guidance released on blackboard
- Report due at the last session
- Present in the last session
  - Group score by instructor, TA, and peers
  - Individual score by peer-evaluation-weighted group score
- AI Yes

Mini Project

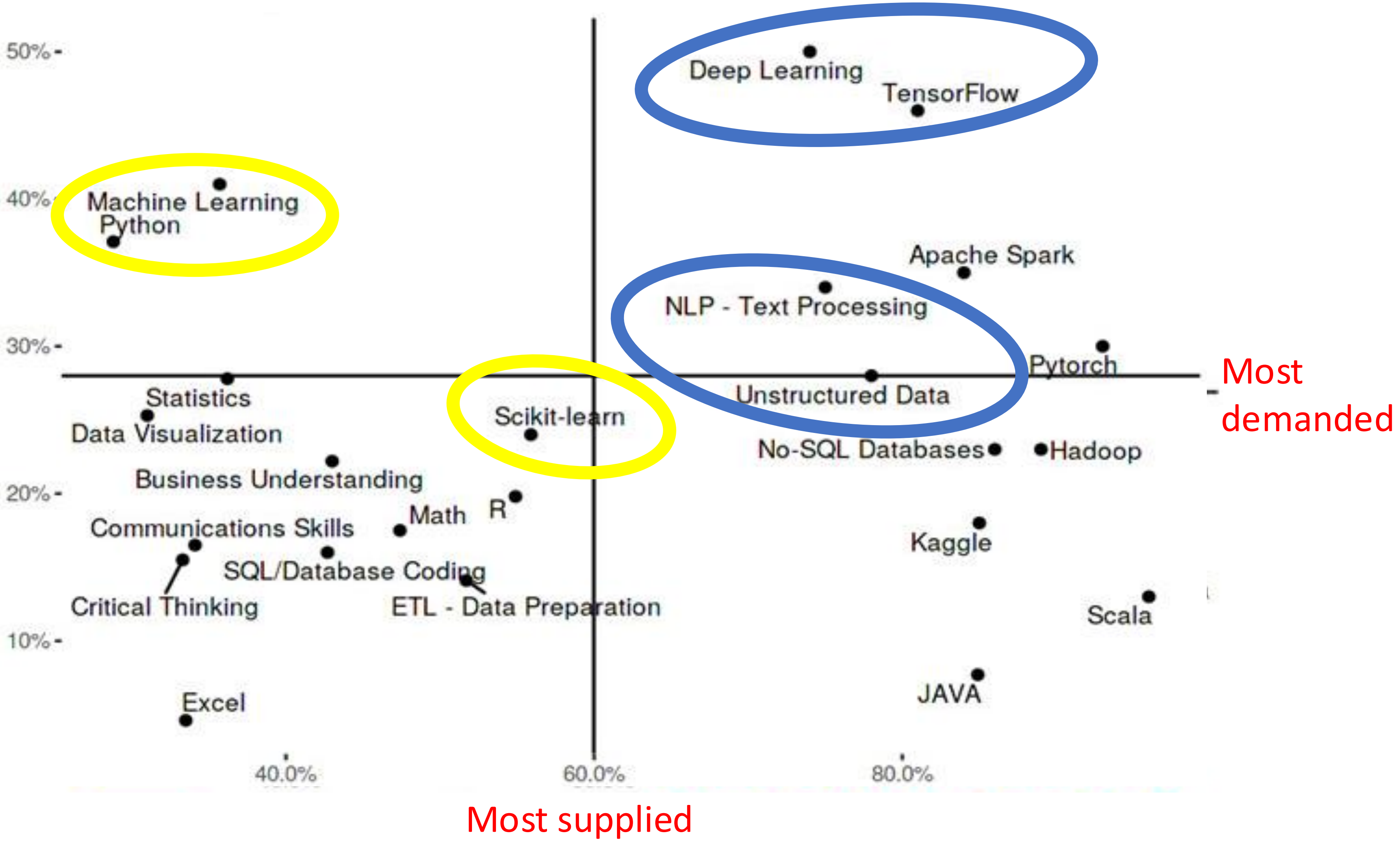
- Every session
- 30 pt (2 points each)
- 4-5 members, shuffling every week
- Topic released during class
- For regular meetings: Usually finish in class, due at the end (23:59) of the next day to the class, no make-up, no late submission
- For asynchronous meetings: due at the end of the next week
- AI Yes

Participation (10pt)

- Warm-up quizzes (the grades are based on participation but not correctness, no late quizzes)
- Q&A and practices, class/group discussions
- This is a hands-on class. I expect many of you will volunteer to share your solutions but if we don't have any volunteers, I will warm/cold call

Your expectation

- Grading: reward for the more technical course
- Make you a (20+80)% BA
  - Analyze social networks
  - Analyze unstructured data
  - Develop AI models
- MOST-WANTED SKILLS



Most popular python tools

Reminder

- Before next meet
  - Carefully review your syllabus and project guidance
  - Install Python, PyCharm and Scrapy (if not Macbook users, see follow-up email)
  - Finish self-intro survey
- Questions?


MELIORA